

Trace Script

Using set -x inside script

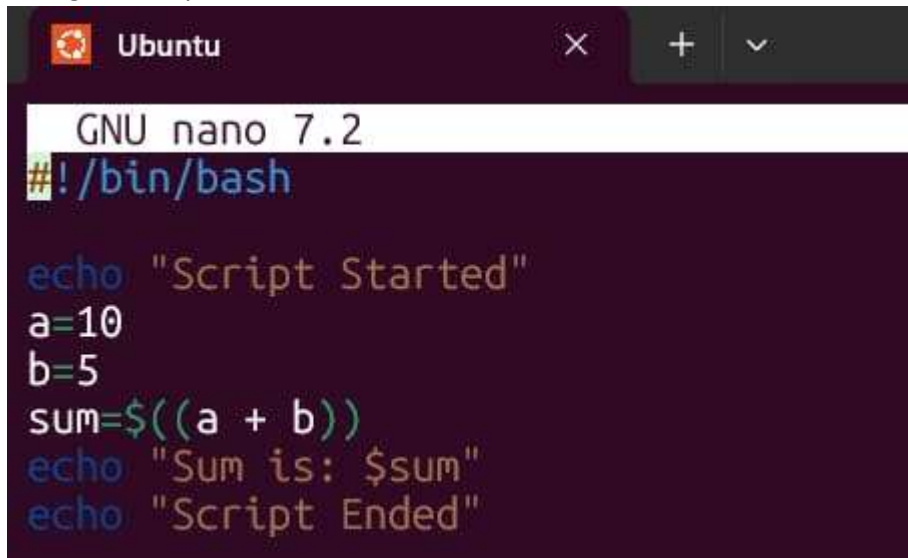
Insert Logging (using echo) inside script



A terminal window titled 'Ubuntu' with standard window controls (close, maximize, minimize). The terminal shows a user named 'priyansh' at a machine named 'IDEAPAD-18UT' in their home directory. They create a file 'debug.sh' using 'nano', set permissions with 'chmod +x debug.sh', and execute it with 'bash -x debug.sh'. The script contains several 'echo' statements and variable assignments. The first execution shows the script running with debugging, outputting 'Script Started', 'Sum is: 15', and 'Script Ended'. The user then runs 'nano debug.sh' again, followed by './debug.sh'. This second execution also shows debugging output, including 'Script Started', 'Sum is: 15', and 'Debugging off'. Finally, the user runs 'nano debug.sh' a third time, followed by './debug.sh'. This third execution shows 'Script Started', 'Sum is: 15', 'Debugging off', and then displays the current user ('Current User is priyansh') and directory ('Current directory is /home/priyansh') before returning to the prompt.

```
priyansh@IDEAPAD-18UT:~$ nano debug.sh
priyansh@IDEAPAD-18UT:~$ chmod +x debug.sh
priyansh@IDEAPAD-18UT:~$ bash -x debug.sh
+ echo 'Script Started'
Script Started
+ a=10
+ b=5
+ sum=15
+ echo 'Sum is: 15'
Sum is: 15
+ echo 'Script Ended'
Script Ended
priyansh@IDEAPAD-18UT:~$
priyansh@IDEAPAD-18UT:~$
priyansh@IDEAPAD-18UT:~$ nano debug.sh
priyansh@IDEAPAD-18UT:~$ nano debug.sh
priyansh@IDEAPAD-18UT:~$ ./debug.sh
+ echo 'Script Started'
Script Started
+ a=10
+ b=5
+ sum=15
+ echo 'Sum is: 15'
Sum is: 15
+ set +x
Debugging off
priyansh@IDEAPAD-18UT:~$ nano debug.sh
priyansh@IDEAPAD-18UT:~$ ./debug.sh
+ echo 'Script Started'
Script Started
+ a=10
+ b=5
+ sum=15
+ echo 'Sum is: 15'
Sum is: 15
+ set +x
Debugging off
Current User is priyansh
Current directory is /home/priyansh
priyansh@IDEAPAD-18UT:~$
```

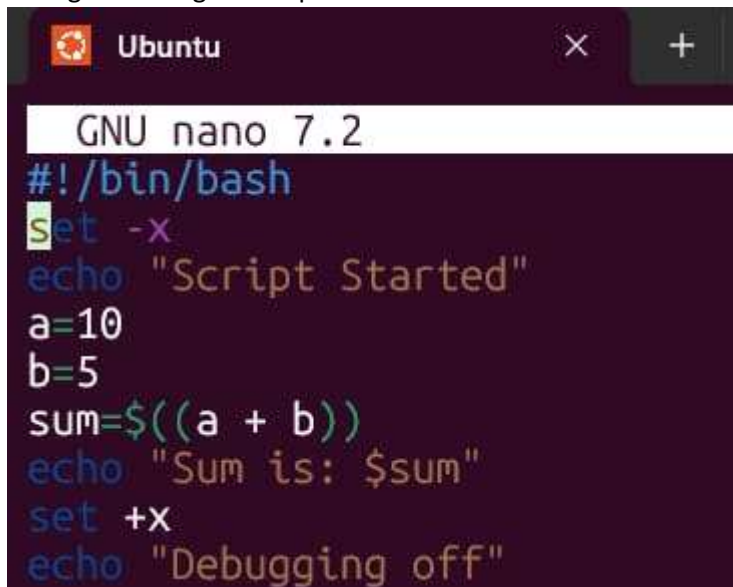
debug.sh – script

A screenshot of a terminal window titled 'Ubuntu'. The terminal shows the GNU nano 7.2 editor with a script. The script starts with a shebang line, followed by echo statements for 'Script Started' and 'Script Ended', and a calculation of the sum of a=10 and b=5.

```
GNU nano 7.2
#!/bin/bash

echo "Script Started"
a=10
b=5
sum=$((a + b))
echo "Sum is: $sum"
echo "Script Ended"
```

debug.sh – using -x in script

A screenshot of a terminal window titled 'Ubuntu'. The terminal shows the GNU nano 7.2 editor with a script that includes debugging. The script starts with a shebang line, followed by 'set -x' to enable debugging, then the same echo statements and calculation as the first script, and finally 'set +x' to disable debugging and an echo statement for 'Debugging off'.

```
GNU nano 7.2
#!/bin/bash
set -x
echo "Script Started"
a=10
b=5
sum=$((a + b))
echo "Sum is: $sum"
set +x
echo "Debugging off"
```

debug.sh – logging using echo

```
Ubuntu
GNU nano 7.2
#!/bin/bash
set -x
echo "Script Started"
a=10
b=5
sum=$((a + b))
echo "Sum is: $sum"
set +x
echo "Debugging off"
echo "Current User is $USER"
echo "Current directory is $(pwd)"
```

time ./ - Measure script runtime

```
priyansh@IDEAPAD-18UT:~$ time ./debug.sh
+ echo 'Script Started'
Script Started
+ a=10
+ b=5
+ sum=15
+ echo 'Sum is: 15'
Sum is: 15
+ set +x
Debugging off
Current User is priyansh
Current directory is /home/priyansh

real    0m0.007s
user    0m0.004s
sys     0m0.001s
priyansh@IDEAPAD-18UT:~$
```

Top Command – to check all processes

```
priyansh@IDEAPAD-18UT: $ top
top - 22:34:45 up 14 min, 1 user, load average: 0.05, 0.14, 0.21
Tasks: 22 total, 1 running, 21 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 1.5 sy, 0.0 ni, 98.5 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 4857.2 total, 4445.1 free, 432.0 used, 103.4 buff/cache
MiB Swap: 2048.0 total, 2048.0 free, 0.0 used, 4425.2 avail Mem

  PID USER      PR  NI   VIRT    RES    SHR   %CPU  %MEM   TIME+ COMMAND
 246 root        20   0   3144    1288    1152   3.0   0.0   0:00.18 /init
 594 priyansh   20   0   9292    5504    3328   3.0   0.1   0:00.27 top
    1 root        20   0  21844   12516   9316   0.0   0.3   0:07.01 /sbin/init
    2 root        20   0   3120    2048    1920   0.0   0.0   0:00.18 /init
    9 root        20   0   3120    1792    1792   0.0   0.0   0:00.00 plan9 --control-socket 7 --log-level 4 --server-fd 8 --pipe-fd 10 --log-truncate
   60 root       19  -1  50432   15616   14720   0.0   0.3   0:01.39 /usr/lib/systemd/systemd-journald
  118 root       20   0  25380    6400    4864   0.0   0.1   0:01.20 /usr/lib/systemd/systemd-udev
  134 systemd+   20   0  21460   12416   10368   0.0   0.2   0:01.28 /usr/lib/systemd/systemd-resolved
  135 systemd+   20   0   91028    7680    6784   0.0   0.2   0:00.85 /usr/lib/systemd/systemd-timesyncd
  191 root       20   0   4236    2432    2304   0.0   0.0   0:00.04 /usr/sbin/cron -f -P
  192 message+   20   0   9552    4736    4352   0.0   0.1   0:00.36 @dbus-daemon -system --address=systemd: --nofork --nopidfile --systemd-activation +
  202 root       20   0  18020    8320    7424   0.0   0.2   0:00.56 /usr/lib/systemd/systemd-logind
  206 syslog     20   0  222508    5248    4352   0.0   0.1   0:00.47 /usr/sbin/rsyslogd -n -iNONE
  214 root       20   0   3160    1920    1792   0.0   0.0   0:00.15 /sbin/agetty -o -p -- \u --noclear --keep-baud - 115200,38400,9600 vt220
  222 root       20   0   3116    1792    1664   0.0   0.0   0:00.09 /sbin/agetty -o -p -- \u --noclear - linux
  229 root       20   0 107024   22272   13056   0.0   0.4   0:01.04 /usr/bin/python3 /usr/share/unattended-upgrades/unattended-upgrade-shutdown --wait+
  245 root       20   0   3128     900     768   0.0   0.0   0:00.01 /init
  253 priyansh   20   0   6072    4992    3456   0.0   0.1   0:00.41 -bash
  254 root       20   0   6692    4224    3584   0.0   0.1   0:00.13 /bin/login -f
  324 priyansh   20   0  20116   10752    8960   0.0   0.2   0:01.50 /usr/lib/systemd/systemd --user
  325 priyansh   20   0  21160    3520    1792   0.0   0.1   0:00.00 (sd-pam)
  383 priyansh   20   0   6072    4864    3328   0.0   0.1   0:00.31 -bash
```

Install Strace

```
priyansh@IDEAPAD-18UT: $ sudo apt install strace
[sudo] password for priyansh:
Sorry, try again.
[sudo] password for priyansh:
Sorry, try again.
[sudo] password for priyansh:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following package was automatically installed and is no longer required:
  libllvm19
Use 'sudo apt autoremove' to remove it.
The following NEW packages will be installed:
  strace
0 upgraded, 1 newly installed, 0 to remove and 0 not upgraded.
Need to get 584 kB of archives.
After this operation, 2153 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble/main amd64 strace amd64 6.8-0ubuntu2 [584 kB]
Fetched 584 kB in 3s (168 kB/s)
Selecting previously unselected package strace.
(Reading database ... 40775 files and directories currently installed.)
Preparing to unpack .../strace_6.8-0ubuntu2_amd64.deb ...
Unpacking strace (6.8-0ubuntu2) ...
Setting up strace (6.8-0ubuntu2) ...
Processing triggers for man-db (2.12.0-4build2) ...
priyansh@IDEAPAD-18UT: $
```

Strace Command Output -

```
rt_sigprocmask(SIG_BLOCK, NULL, [], 0) = 0
rt_sigaction(SIGCHLD, {sa_handler=SIG_DFL, sa_mask=[], sa_flags=SA_RESTORER|SA_RESTART, sa_restorer=0x720ca3445330}, {sa_handler=SIG_DFL, sa_mask=[], sa_flags=0}, 0) = 0
rt_sigaction(SIGCHLD, {sa_handler=SIG_DFL, sa_mask=[], sa_flags=SA_RESTORER|SA_RESTART, sa_restorer=0x720ca3445330}, {sa_handler=SIG_DFL, sa_mask=[], sa_flags=SA_RESTORER|SA_RESTART, sa_restorer=0x720ca3445330}, 0) = 0
rt_sigaction(SIGINT, {sa_handler=SIG_DFL, sa_mask=[], sa_flags=SA_RESTORER, sa_restorer=0x720ca3445330}, {sa_handler=SIG_DFL, sa_mask=[], sa_flags=0}, 0) = 0
rt_sigaction(SIGINT, {sa_handler=SIG_DFL, sa_mask=[], sa_flags=SA_RESTORER, sa_restorer=0x720ca3445330}, {sa_handler=SIG_DFL, sa_mask=[], sa_flags=SA_RESTORER, sa_restorer=0x720ca3445330}, 0) = 0
rt_sigaction(SIGQUIT, {sa_handler=SIG_DFL, sa_mask=[], sa_flags=SA_RESTORER, sa_restorer=0x720ca3445330}, {sa_handler=SIG_DFL, sa_mask=[], sa_flags=0}, 0) = 0
rt_sigaction(SIGQUIT, {sa_handler=SIG_DFL, sa_mask=[], sa_flags=SA_RESTORER, sa_restorer=0x720ca3445330}, {sa_handler=SIG_DFL, sa_mask=[], sa_flags=SA_RESTORER, sa_restorer=0x720ca3445330}, 0) = 0
rt_sigaction(SIGSTP, {sa_handler=SIG_DFL, sa_mask=[], sa_flags=SA_RESTORER, sa_restorer=0x720ca3445330}, {sa_handler=SIG_DFL, sa_mask=[], sa_flags=0}, 0) = 0
rt_sigaction(SIGSTP, {sa_handler=SIG_DFL, sa_mask=[], sa_flags=SA_RESTORER, sa_restorer=0x720ca3445330}, {sa_handler=SIG_DFL, sa_mask=[], sa_flags=SA_RESTORER, sa_restorer=0x720ca3445330}, 0) = 0
rt_sigaction(SIGTTIN, {sa_handler=SIG_DFL, sa_mask=[], sa_flags=SA_RESTORER, sa_restorer=0x720ca3445330}, {sa_handler=SIG_DFL, sa_mask=[], sa_flags=0}, 0) = 0
rt_sigaction(SIGTTIN, {sa_handler=SIG_DFL, sa_mask=[], sa_flags=SA_RESTORER, sa_restorer=0x720ca3445330}, {sa_handler=SIG_DFL, sa_mask=[], sa_flags=SA_RESTORER, sa_restorer=0x720ca3445330}, 0) = 0
rt_sigaction(SIGTTOU, {sa_handler=SIG_DFL, sa_mask=[], sa_flags=SA_RESTORER, sa_restorer=0x720ca3445330}, {sa_handler=SIG_DFL, sa_mask=[], sa_flags=0}, 0) = 0
rt_sigaction(SIGTTOU, {sa_handler=SIG_DFL, sa_mask=[], sa_flags=SA_RESTORER, sa_restorer=0x720ca3445330}, {sa_handler=SIG_DFL, sa_mask=[], sa_flags=SA_RESTORER, sa_restorer=0x720ca3445330}, 0) = 0
rt_sigprocmask(SIG_BLOCK, NULL, [], 0) = 0
rt_sigaction(SIG_IGN, {sa_handler=SIG_IGN, sa_mask=[], sa_flags=SA_RESTORER, sa_restorer=0x720ca3445330}, {sa_handler=SIG_DFL, sa_mask=[], sa_flags=SA_RESTORER, sa_restorer=0x720ca3445330}, 0) = 0
uname({sysname="Linux", nodename="IDEAPAD-18U7", ...}) = 0
newfstatat(AT_FDCWD, "/home/priyansh", {st_mode=S_IFDIR|0750, st_size=4096, ...}, 0) = 0
newfstatat(AT_FDCWD, ".", {st_mode=S_IFDIR|0750, st_size=4096, ...}, 0) = 0
newfstatat(AT_FDCWD, "/home", {st_mode=S_IFDIR|0755, st_size=4096, ...}, 0) = 0
newfstatat(AT_FDCWD, "/home/priyansh", {st_mode=S_IFDIR|0750, st_size=4096, ...}, 0) = 0
getpid()
= 704
getpid()
= 701
getpid()
= 704
getpid()
= 701
getpid()
= 704
getpid()
= 701
getpid()
= 701
ioctl(2, TIOCCGRP, [701])
= 0
rt_sigaction(SIGCHLD, {sa_handler=0x608601dc4230, sa_mask=[], sa_flags=SA_RESTORER|SA_RESTART, sa_restorer=0x720ca3445330}, {sa_handler=SIG_DFL, sa_mask=[], sa_flags=SA_RESTORER|SA_RESTART, sa_restorer=0x720ca3445330}, 0) = 0
prlimit64(0, RLIMIT_NPROC, NULL, {rlim_cur=19389, rlim_max=19389}) = 0
rt_sigprocmask(SIG_BLOCK, NULL, [], 0) = 0
openat(AT_FDCWD, "./debug.sh", O_RDONLY) = 3
newfstatat(AT_FDCWD, "./debug.sh", {st_mode=S_IFREG|0755, st_size=185, ...}, 0) = 0
ioctl(3, TCGETS, 0x7ff62727afcd) = -1 ENOTTY (Inappropriate ioctl for device)
lseek(3, 0, SEEK_CUR)
= 0
read(3, "#!/bin/bash\nset -x\n\nScript '...', 80) = 80
lseek(3, 0, SEEK_SET)
= 0
prlimit64(0, RLIMIT_NOFILE, NULL, {rlim_cur=10*1024, rlim_max=1024*1024}) = 0
fcntl(255, F_GETFD)
= -1 EBADF (Bad file descriptor)
dup2(3, 255)
= 255
close(3)
= 0
fcntl(255, F_SETFD, FD_CLOEXEC)
= 0
fcntl(255, F_GETFL)
= 0x8000 (Flags O_RDONLY|O_LARGEFILE)
fstat(255, {st_mode=S_IFREG|0755, st_size=185, ...}) = 0
lseek(255, 0, SEEK_CUR)
= 0
read(255, "#!/bin/bash\nset -x\n\nScript '...', 185) = 185
fstat(2, {st_mode=S_IFCHR|0620, st_rdev=makedev(0x88, 0), ...}) = 0
write(2, "+ echo 'Script Started'\n", 24) echo 'Script Started'
) = 24
fstat(1, {st_mode=S_IFCHR|0620, st_rdev=makedev(0x88, 0), ...}) = 0
write(1, "Script Started\n", 15)Script Started
)
= 15
write(2, "+ a=10\n", 7) a=10
)
= 7
write(2, "+ b=5\n", 6) b=5
)
= 6
write(2, "+ sum=15\n", 9) sum=15
)
= 9
write(2, "+ echo 'Sum is: 15'\n", 20) echo 'Sum is: 15'
)
= 20
write(1, "Sum is: 15\n", 11)Sum is: 15
)
= 11
write(2, "+ set -x\n", 9) set -x
)
= 9
```



```
prlyansh@IDEAPAD-18UT: $ strace ./debug.sh
execve("./debug.sh", ["/debug.sh"], 0x7ffcf9e7330 /* 27 vars */) = 0
brk(NULL)                               = 0x6086e9a1f000
mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x720ca3794000
access("/etc/ld.so.preload", R_OK)      = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=17575, ...}) = 0
mmap(NULL, 17575, PROT_READ, MAP_PRIVATE, 3, 0) = 0x720ca378f000
close(3)                                 = 0
openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libtinfo.so.6", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\0\0\0\0\0>\0\1\0\0\0\0\0\0\0\0\0\0\0...", 832) = 832
fstat(3, {st_mode=S_IFREG|0644, st_size=208328, ...}) = 0
mmap(NULL, 211240, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x720ca375b000
mmap(0x720ca3769000, 77824, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0xe000) = 0x720ca3769000
mmap(0x720ca377c000, 57344, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x21000) = 0x720ca377c000
mmap(0x720ca378a000, 20480, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x2e000) = 0x720ca378a000
close(3)                                 = 0
openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libc.so.6", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\0\0\0\0\0>\0\1\0\0\0\0\0\0\0\0\0\0\0...", 832) = 832
pread64(3, "\6\0\0\0\4\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0...", 784, 64) = 784
fstat(3, {st_mode=S_IFREG|0755, st_size=2125328, ...}) = 0
pread64(3, "\6\0\0\0\4\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0...", 784, 64) = 784
mmap(NULL, 2170256, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x720ca3400000
mmap(0x720ca3428000, 1605632, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x28000) = 0x720ca3428000
mmap(0x720ca35b0000, 323584, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x1b0000) = 0x720ca35b0000
mmap(0x720ca35ff000, 24576, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x1fe000) = 0x720ca35ff000
mmap(0x720ca3605000, 52624, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x720ca3605000
close(3)                                 = 0
mmap(NULL, 12288, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x720ca3758000
arch_prctl(ARCH_SET_FS, 0x720ca3758740) = 0
set_tid_address(0x720ca3758a10)          = 704
set_robust_list(0x720ca3758a20, 24)     = 0
rseq(0x720ca3759060, 0x20, 0, 0x53053053) = 0
mprotect(0x720ca35ff000, 16384, PROT_READ) = 0
mprotect(0x720ca378a000, 16384, PROT_READ) = 0
mprotect(0x6086d1eb2000, 16384, PROT_READ) = 0
mprotect(0x720ca37cc000, 8192, PROT_READ) = 0
prlimit64(0, RLIMIT_STACK, NULL, {rlim_cur=8192*1024, rlim_max=RLIM64_INFINITY}) = 0
munmap(0x720ca378f000, 17575)           = 0
openat(AT_FDCWD, "/dev/tty", O_RDWR|O_NONBLOCK) = 3
close(3)                                 = 0
getrandom("\x82\x0b\x9a\x70\x99\x24\x76\x3f", 8, GRND_NONBLOCK) = 8
brk(NULL)                               = 0x6086e9a1f000
brk(0x6086e9a40000)                     = 0x6086e9a40000
openat(AT_FDCWD, "/usr/lib/locale/locale-archive", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/usr/share/locale/locale.alias", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=2996, ...}) = 0
read(3, "# Locale name alias data base.\n#...", 4096) = 2996
read(3, "", 4096)                       = 0
close(3)                                 = 0
openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_IDENTIFICATION", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_IDENTIFICATION", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=258, ...}) = 0
mmap(NULL, 258, PROT_READ, MAP_PRIVATE, 3, 0) = 0x720ca3793000
close(3)                                 = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/gconv/gconv-modules.cache", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=27028, ...}) = 0
mmap(NULL, 27028, PROT_READ, MAP_SHARED, 3, 0) = 0x720ca3751000
close(3)                                 = 0
futext(0x720ca360472c, FUTEX_WAKE_PRIVATE, 2147483647) = 0
openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_MEASUREMENT", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_MEASUREMENT", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=23, ...}) = 0
mmap(NULL, 23, PROT_READ, MAP_PRIVATE, 3, 0) = 0x720ca3792000
```

```
Ubuntu X + v

mmap(NULL, 23, PROT_READ, MAP_PRIVATE, 3, 0) = 0x728ca3792000
close(3)
= 0
openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_TELEPHONE", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_TELEPHONE", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG(0644), st_size=47, ...}) = 0
mmap(NULL, 47, PROT_READ, MAP_PRIVATE, 3, 0) = 0x728ca3791000
close(3)
= 0
openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_ADDRESS", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_ADDRESS", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG(0644), st_size=127, ...}) = 0
mmap(NULL, 127, PROT_READ, MAP_PRIVATE, 3, 0) = 0x728ca3790000
close(3)
= 0
openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_NAME", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_NAME", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG(0644), st_size=62, ...}) = 0
mmap(NULL, 62, PROT_READ, MAP_PRIVATE, 3, 0) = 0x728ca378f000
close(3)
= 0
openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_PAPER", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_PAPER", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG(0644), st_size=34, ...}) = 0
mmap(NULL, 34, PROT_READ, MAP_PRIVATE, 3, 0) = 0x728ca378e000
close(3)
= 0
openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_MESSAGES", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_MESSAGES", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFDIR(0755), st_size=4096, ...}) = 0
close(3)
openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_MESSAGES/SYS_LC_MESSAGES", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG(0644), st_size=48, ...}) = 0
mmap(NULL, 48, PROT_READ, MAP_PRIVATE, 3, 0) = 0x728ca374e000
close(3)
= 0
openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_MONETARY", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_MONETARY", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG(0644), st_size=270, ...}) = 0
mmap(NULL, 270, PROT_READ, MAP_PRIVATE, 3, 0) = 0x728ca374d000
close(3)
= 0
openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_COLLATE", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_COLLATE", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG(0644), st_size=1406, ...}) = 0
mmap(NULL, 1406, PROT_READ, MAP_PRIVATE, 3, 0) = 0x728ca374c000
close(3)
= 0
openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_TIME", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_TIME", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG(0644), st_size=3360, ...}) = 0
mmap(NULL, 3360, PROT_READ, MAP_PRIVATE, 3, 0) = 0x728ca374c000
close(3)
= 0
openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_NUMERIC", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_NUMERIC", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG(0644), st_size=50, ...}) = 0
mmap(NULL, 50, PROT_READ, MAP_PRIVATE, 3, 0) = 0x728ca374b000
close(3)
= 0
openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_CTYPE", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_CTYPE", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG(0644), st_size=360460, ...}) = 0
mmap(NULL, 360460, PROT_READ, MAP_PRIVATE, 3, 0) = 0x728ca36f2000
close(3)
= 0
getuid()
= 1000
getgid()
= 1000
geteuid()
= 1000
getegid()
= 1000
rt_sigprocmask(SIG_BLOCK, NULL, [], 0)
= 0
rt_sigaction(SIGCHLD, {sa_handler=SIG_DFL, sa_mask=[], sa_flags=SA_RESTORER|SA_RESTART, sa_restorer=0x728ca3445330}, {sa_handler=SIG_DFL, sa_mask=[], sa_flags=0}, 0) = 0
rt_sigaction(SIGCHLD, {sa_handler=SIG_DFL, sa_mask=[], sa_flags=SA_RESTORER|SA_RESTART, sa_restorer=0x728ca3445330}, {sa_handler=SIG_DFL, sa_mask=[], sa_flags=SA_RESTORER|SA_RESTART, sa_restorer=0x728ca3445330}, 0) = 0
rt_sigaction(SIGINT, {sa_handler=SIG_DFL, sa_mask=[], sa_flags=SA_RESTORER, sa_restorer=0x728ca3445330}, {sa_handler=SIG_DFL, sa_mask=[], sa_flags=0}, 0) = 0
rt_sigaction(SIGINT, {sa_handler=SIG_DFL, sa_mask=[], sa_flags=SA_RESTORER, sa_restorer=0x728ca3445330}, {sa_handler=SIG_DFL, sa_mask=[], sa_flags=SA_RESTORER, sa_restorer=0x728ca3445330}, 0) = 0

Ubuntu X + v

write(2, "+ set +x\n", 9+ set +x
)
= 9
write(1, "Debugging off\n", 14Debugging off
)
= 14
write(1, "Current User is priyansh\n", 25Current User is priyansh
)
= 25
rt_sigprocmask(SIG_BLOCK, NULL, [], 0) = 0
rt_sigprocmask(SIG_BLOCK, NULL, [], 0) = 0
pipe2([3, 4], 0)
= 0
rt_sigprocmask(SIG_BLOCK, [CHLD], [], 0) = 0
rt_sigprocmask(SIG_SETMASK, [], NULL, 0) = 0
rt_sigprocmask(SIG_BLOCK, NULL, [], 0) = 0
rt_sigprocmask(SIG_BLOCK, [INT TERM CHLD], [], 0) = 0
lseek(255, -8, SEEK_CUR)
= 177
clone(child_stack=NULL, flags=CLONE_CHILD_CLEARTID|CLONE_CHILD_SETTID|SIGCHLD, child_tidptr=0x728ca3758a10) = 705
rt_sigprocmask(SIG_SETMASK, [], NULL, 0) = 0
rt_sigaction(SIGCHLD, {sa_handler=0x6086d1dc4230, sa_mask=[], sa_flags=SA_RESTORER|SA_RESTART, sa_restorer=0x728ca3445330}, {sa_handler=SIG_DFL, sa_mask=[], sa_flags=SA_RESTORER|SA_RESTART, sa_restorer=0x728ca3445330}, 0) = 0
close(4)
... SIGCHLD (si_signo=SIGCHLD, si_code=CLD_EXITED, si_pid=705, si_uid=1000, si_status=0, si_utime=0, si_stime=0) ...
wait4(-1, [WIFEXITED(s) && WEXITSTATUS(s) == 0], WNOHANG, NULL) = 705
wait4(-1, 0x7ffe2727a210, WNOHANG, NULL) = -1 ECHILD (No child processes)
rt_sigreturn(mask=[])
= 0
rt_sigprocmask(SIG_BLOCK, [INT], [], 0) = 0
read(3, "/home/priyansh\n", 4096)
= 15
read(3, "", 4096)
= 0
close(3)
= 0
rt_sigprocmask(SIG_SETMASK, [], NULL, 0) = 0
rt_sigprocmask(SIG_BLOCK, [CHLD], [], 0) = 0
rt_sigaction(SIGINT, {sa_handler=0x6086d1dc6750, sa_mask=[], sa_flags=SA_RESTORER, sa_restorer=0x728ca3445330}, {sa_handler=SIG_DFL, sa_mask=[], sa_flags=SA_RESTORER, sa_restorer=0x728ca3445330}, 0) = 0
rt_sigaction(SIGINT, {sa_handler=SIG_DFL, sa_mask=[], sa_flags=SA_RESTORER, sa_restorer=0x728ca3445330}, {sa_handler=0x6086d1dc6750, sa_mask=[], sa_flags=SA_RESTORER, sa_restorer=0x728ca3445330}, 0) = 0
rt_sigprocmask(SIG_SETMASK, [], NULL, 0) = 0
write(1, "Current directory is /home/priya...., 36Current directory is /home/priyansh
)
= 36
read(255, "\n\n\n\n\n\n\n\n\n", 105)
= 0
read(255, "", 105)
= 0
rt_sigprocmask(SIG_BLOCK, [CHLD], [], 0) = 0
rt_sigprocmask(SIG_SETMASK, [], NULL, 0) = 0
exit_group(0)
= ?
+++ exited with 0 +++
priyansh@IDEAPAD-18UT: $
```

Install Linux Command Tools to use stat command

```
Ubuntu
priyansh@IDEAPAD-18UT:~$ sudo apt install linux-laptop-tools-common
Waiting for cache lock: Could not get lock /var/lib/dpkg/lock-frontent. It is held by process 1067 (unattended-upgr)
Waiting for cache lock: Could not get lock /var/lib/dpkg/lock-frontent. It is held by process 1067 (unattended-upgr)
Waiting for cache lock: Could not get lock /var/lib/dpkg/lock-frontent. It is held by process 1067 (unattended-upgr)
Waiting for cache lock: Could not get lock /var/lib/dpkg/lock-frontent. It is held by process 1067 (unattended-upgr)
Waiting for cache lock: Could not get lock /var/lib/dpkg/lock-frontent. It is held by process 1067 (unattended-upgr)
Waiting for cache lock: Could not get lock /var/lib/dpkg/lock-frontent. It is held by process 1067 (unattended-upgr)
^Citing for cache lock: Could not get lock /var/lib/dpkg/lock-frontent. It is held by process 1067 (unattended-upgr)... 7s
priyansh@IDEAPAD-18UT:~$ sudo apt install linux-tools-common
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following package was automatically installed and is no longer required:
  liblvm19
Use 'sudo apt autoremove' to remove it.
The following additional packages will be installed:
  hwdata
The following NEW packages will be installed:
  hwdata linux-tools-common
0 upgraded, 2 newly installed, 0 to remove and 67 not upgraded.
Need to get 964 kB of archives.
After this operation, 1118 kB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://archive.ubuntu.com/ubuntu noble/main amd64 hwdata all 0.379-1 [29.1 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 linux-tools-common all 6.8.0-106.106 [935 kB]
Fetched 964 kB in 4s (231 kB/s)
Selecting previously unselected package hwdata.
(Reading database ... 40785 files and directories currently installed.)
Preparing to unpack .../hwdata_0.379-1_all.deb ...
Unpacking hwdata (0.379-1) ...
Selecting previously unselected package linux-tools-common.
Preparing to unpack .../linux-tools-common_6.8.0-106.106_all.deb ...
Unpacking linux-tools-common (6.8.0-106.106) ...
Setting up hwdata (0.379-1) ...
Setting up linux-tools-common (6.8.0-106.106) ...
Processing triggers for man-db (2.12.0-4build2) ...
priyansh@IDEAPAD-18UT:~$
```

perf stat – to check details/statistics

```
priyansh@IDEAPAD-18UT:~$ ls /usr/lib/linux-tools/
6.8.0-106-generic
priyansh@IDEAPAD-18UT:~$ /usr/lib/linux-tools/6.8.0-106-generic/perf stat ./debug.sh
+ echo 'Script Started'
Script Started
+ a=10
+ b=5
+ sum=15
+ echo 'Sum is: 15'
Sum is: 15
+ set +x
Debugging off
Current User is priyansh
Current directory is /home/priyansh

Performance counter stats for './debug.sh':

    12.08 msec task-clock                    #    0.796 CPUs utilized
         3      context-switches            #   248.339 /sec
         0      cpu-migrations              #    0.000 /sec
        231     page-faults                 #   19.122 K/sec
   12629056     cycles                      #    1.045 GHz
   1539168     stalled-cycles-frontend     #   12.19% frontend cycles idle
   1476113     stalled-cycles-backend      #   11.69% backend cycles idle
   5540958     instructions                 #    0.44 insns per cycle
                                   #   0.28 stalled cycles per insn
   1250350     branches                    #  103.504 M/sec
    167051     branch-misses                #   13.36% of all branches

    0.015175268 seconds time elapsed

    0.003830000 seconds user
    0.009634000 seconds sys

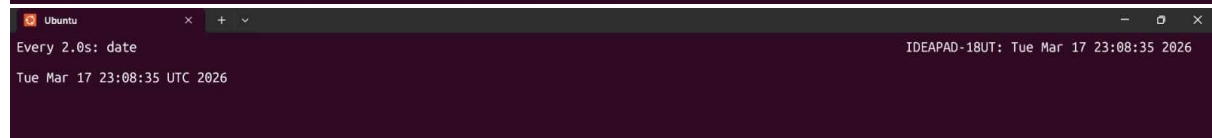
priyansh@IDEAPAD-18UT:~$
```

Using find, xargs, tee commands

```
priyansh@IDEAPAD-18UT:~$ find . -name "*.txt"
./output.txt
priyansh@IDEAPAD-18UT:~$ find . -name "*.txt" | xargs wc -l
wc: invalid option -- '1'
Try 'wc --help' for more information.
priyansh@IDEAPAD-18UT:~$ find . -name "*.txt" | xargs wc -l
3 ./output.txt
priyansh@IDEAPAD-18UT:~$ ls | tee output.txt
debug.sh
nohup.out
output.txt
priyansh@IDEAPAD-18UT:~$
```

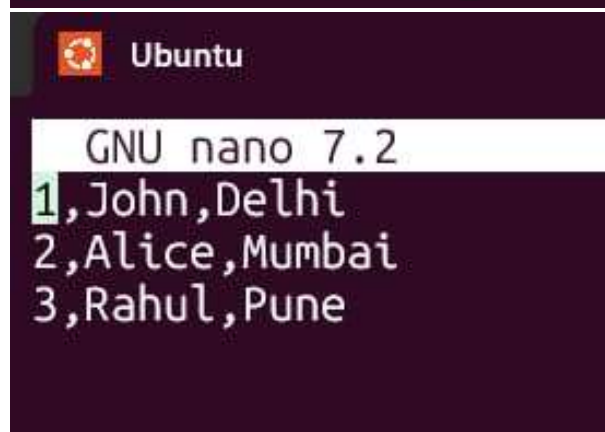
Using tr, yes, watch commands (ctrl + c – to exit watch date view)

```
priyansh@IDEAPAD-18UT:~$ echo "linux" | tr 'a-z' 'A-Z'
LINUX
priyansh@IDEAPAD-18UT:~$ yes "HELLO" | head -n 5
HELLO
HELLO
HELLO
HELLO
HELLO
priyansh@IDEAPAD-18UT:~$ watch date
priyansh@IDEAPAD-18UT:~$
```

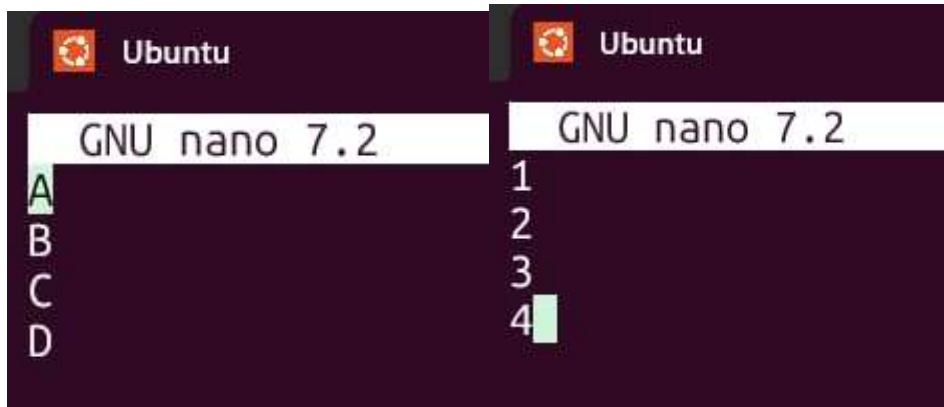


Create data.csv file and use cut command on it to remove any character/number/word from the file

```
priyansh@IDEAPAD-18UT:~$ nano data.csv
priyansh@IDEAPAD-18UT:~$ cut -d',' -f2 data.csv
John
Alice
Rahul
priyansh@IDEAPAD-18UT:~$
```



Make 2 files -
file1.txt & file2.txt



```
Ubuntu
GNU nano 7.2
A
B
C
D

Ubuntu
GNU nano 7.2
1
2
3
4
```

paste – to join two files



```
priyansh@IDEAPAD-18UT:~$ nano file1.txt
priyansh@IDEAPAD-18UT:~$ nano file2.txt
priyansh@IDEAPAD-18UT:~$ paste file1.txt file2.txt
A      1
B      2
C      3
D      4
priyansh@IDEAPAD-18UT:~$
```

Pipeline

Rename Files in Batch

Extract Columns from CSV



```
priyansh@IDEAPAD-18UT:~$ ls | grep ".txt" | wc -l
wc: invalid option -- '1'
Try 'wc --help' for more information.
priyansh@IDEAPAD-18UT:~$ ls | grep ".txt" | wc -l
3
priyansh@IDEAPAD-18UT:~$ touch file1.txt file2.txt file3.txt
priyansh@IDEAPAD-18UT:~$ for f in *.txt
> do
> mv "$f" "new_$f"
> done
priyansh@IDEAPAD-18UT:~$ ls
data.csv  debug.sh  new_file1.txt  new_file2.txt  new_file3.txt
priyansh@IDEAPAD-18UT:~$
```